THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET A

Chapters 9 & 10: **Light – Reflection and Refraction, The Human Eye and the Colourful World** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 30 cm	$R = 2f$
2	**(a)** $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$	Mirror formula has +
3	**(b)** virtual and erect	Laterally inverted
4	**(a)** Correct	Object between P and F
5	**(a)** −5 D	$1/(-0.2)$
6	**(b)** retina	Screen
7	**(d)** outer surface of the cornea	Cornea
8	**(a)** less than infinity	Nearer
9	**(a)** Incorrect — violet is shorter; our eyes are more sensitive to blue	Scattering
10	**(a)** myopia	Concave lens

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	(i) concave (ii) convex	1+1
12	$m = -2$	2
13	(i) iris/pupil (ii) ciliary muscles	1+1
14	The eye lens becomes cloudy/opaque and does not let light pass properly.	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$\frac{1}{v} = -\frac{1}{10} + \frac{1}{30} = -\frac{1}{15} \Rightarrow v = -15$ cm; $m = -v/u = -0.5 \Rightarrow h' = -1$ cm	2+1
16	Myopia; concave lens; $f = -1.2$ m $\Rightarrow P = -0.83$ D	1+1+1
17	$\frac{1}{v} = \frac{1}{8} - \frac{1}{12} = \frac{1}{24} \Rightarrow v = +24$ cm; real, inverted, $m = -2$	2+1
	OR: Ray parallel to axis passes through F (concave); ray through F becomes parallel; ray through C retraces; ray at pole reflects symmetrically	3
18	Ciliary muscles relax for distant objects (lens thin, long focal length) and contract for near objects (lens thick, short focal length)	3
	OR: Air molecules are smaller than the wavelength of visible light, so they scatter shorter-wavelength blue light more strongly; the scattered blue light enters our eyes	3

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $m = h'/h = -v/u$ (ii) $\frac{1}{v} = -\frac{1}{12} + \frac{1}{6} = \frac{1}{12} \Rightarrow v = +12$ cm (behind mirror); $m = 2$; virtual, erect, magnified	1+4
	OR: (i) 40 cm (2F) (ii) virtual, erect, magnified — no image on screen (iii) $\frac{1}{v} = \frac{1}{20} - \frac{1}{30} = \frac{1}{60} \Rightarrow v = 60$ cm (iv) $m = v/u = -2$ (v) +5 D	1+1+1+1+1
20	(i) cornea, iris, pupil, lens, ciliary muscle, retina, optic nerve with functions (ii) image remains on retina for about 1/16 s after the object is removed	4+1
	OR: (i) scattering of light (ii) air molecules scatter blue more (iii) red is least scattered and visible from far (iv) red (v) fine smoke particles scatter blue light	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET B

Chapters 9 & 10: **Light – Reflection and Refraction, The Human Eye and the Colourful World** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** Student A is correct	Sign convention
2	**(c)** 0°	Along normal
3	**(a)** concave	Converges sunlight
4	**(a)** real, inverted and magnified	Negative m
5	**(c)** less than the angle of incidence	Towards normal
6	**(b)** eye lens becomes milky and cloudy	Surgery restores vision
7	**(b)** Student A is correct	Longer path
8	**(b)** 1/16 s	Persistence of vision
9	**(a)** blue/violet	Short wavelength
10	**(d)** angle of deviation	D

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$f = 1/4$ m = 25 cm; convex lens	1+1
12	False — it is a ratio of two speeds, so it has no unit.	2
13	Different colours have different speeds in glass and bend through different angles.	2
14	False — it is due to scattering of blue light by air molecules.	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	For a given pair of media and colour, $\sin i / \sin r$ is constant (Snell's law); $n = c/v$	3
16	Near point: minimum distance for clear vision (25 cm); far point: farthest point seen clearly (infinity)	3
17	Image at 2F ⇒ $2f = 40$ cm ⇒ $f = 20$ cm; $P = +5$ D	2+1
	OR: Real: formed by actual intersection of rays, can be obtained on screen, inverted; virtual: rays appear to meet, cannot be obtained on screen, erect	3
18	Scattering of light by colloidal particles; sunlight entering a dense forest canopy; beam in a smoke-filled room	3
	OR: Sunrise/sunset: light travels longer through atmosphere — blue scattered away, red reaches; noon: shorter path, little scattering	3

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) incident, refracted ray and normal in same plane; $\sin i / \sin r$ = constant (ii) $n = 3 \times 10^8 / 1.5 \times 10^8 = 2$ (iii) refraction at water surface makes the bottom appear raised	2+2+1
	OR: (i) (a) beyond 2F, real, inverted, magnified (b) same side, virtual, erect, magnified (ii) $v = 20$ cm, $m = -1$, $h' = -3$ cm	3+2
20	(i) myopia, hypermetropia (ii) myopia: elongated eyeball/curved lens → concave lens; hypermetropia: short eyeball/long focal length → convex lens	1+4
	OR: (i) refraction of light by the atmosphere with varying density (ii) (a) changing apparent position and brightness (b) Sun seen before it crosses the horizon (iii) planets are extended sources	1+3+1

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ANSWER KEY & MARKING SCHEME SET C

Chapters 9 & 10: **Light – Reflection and Refraction, The Human Eye and the Colourful World** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** +2 D	Add algebraically
2	**(a)** Correct	Parallel rays meet at F
3	**(c)** v_1/v_2	$n_{21} = v_1/v_2$
4	**(c)** at 2F, same size	Real, inverted
5	**(a)** Student A is correct	Virtual magnified
6	**(a)** convex	Converging
7	**(a)** Correct	Less scattering of blue
8	**(d)** cones in the retina	Cones
9	**(c)** refraction, dispersion and internal reflection	Raindrop as prism
10	**(a)** Student A is correct	Hypermetropia: behind

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	The centre of the sphere of which the reflecting surface is a part.	2
12	It always gives an erect, diminished image and has a wider field of view.	2
13	Wider field of view and three-dimensional (depth) perception.	2
14	It remains the same (on the retina); the eye adjusts its focal length.	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$v = -6$, $f = -10$: $\frac{1}{u} = \frac{1}{v} - \frac{1}{f} = -\frac{1}{6} + \frac{1}{10} = -\frac{1}{15} \Rightarrow u = -15$ cm	3
16	Retina converts light to electrical signals; optic nerve sends them to the brain; rods sense dim light, cones sense colour	3
17	Ray bends towards normal entering glass, away from normal on leaving; emergent ray parallel to incident ray with lateral shift	3
	OR: (i) virtual, erect, magnified (ii) real, inverted, same size (iii) real, inverted, diminished	1+1+1
18	Gradual weakening of ciliary muscles and loss of lens flexibility with age; corrected by bifocal lenses	3
	OR: Starlight passes through layers of changing refractive index; its apparent position and brightness fluctuate	3

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) convex mirror (ii) image is diminished so objects appear farther (iii) $f = +1$ m (iv) $\frac{1}{v} = 1 + \frac{1}{10} = \frac{11}{10} \Rightarrow v = +0.91$ m (v) no, virtual	1+1+1+1+1
	OR: (i) $P = 1/f$ in m, unit dioptre (ii) $P = 4 - 10 = -6$ D (iii) no, diverging	1+3+1
20	(i) dispersion (ii) red top, violet bottom (iii) slowest speed / highest refractive index in glass (iv) rainbow (v) white light is made of seven colours	1+1+1+1+1
	OR: (i) adjustment of focal length (ii) contraction thickens lens for near objects; relaxation thins it for far (iii) $f = -80$ cm, $P = -1.25$ D	1+2+2

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ANSWER KEY & MARKING SCHEME SET D

Chapters 9 & 10: **Light – Reflection and Refraction, The Human Eye and the Colourful World** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** both concave mirror and concave lens	Negative f
2	**(c)** 3×10^8 m/s	c
3	**(a)** passes through F	Converges
4	**(d)** $h'/h = v/u$	Lens
5	**(a)** +4 D	$1/0.25$
6	**(d)** three	Triangular prism
7	**(c)** hypermetropia	Far-sighted
8	**(c)** least scattered	Visible from far
9	**(d)** atmospheric refraction	Bending
10	**(d)** with ageing	Old age

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Perpendicular distance between the incident ray and the emergent ray in a glass slab.	2
12	$v = c/n = 2 \times 10^8$ m/s	2
13	In a slab, the two surfaces are parallel so the colours recombine; in a prism they diverge.	2
14	Dark/black — there is little air to scatter light.	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Face placed between P and F gives an erect, magnified, virtual image	3
16	White light refracts at the first face; each colour bends by a different angle; the colours spread out more at the second face and form a spectrum	3
17	$u = v = -20$: $\frac{1}{f} = -\frac{1}{20} - \frac{1}{20} = -\frac{1}{10} \Rightarrow f = -10$ cm (object at C)	3
	OR: $P = +1.0$ D; $f = 1$ m (100 cm), converging	1½+1½
18	$u = -25$ cm, $v = -100$ cm: $\frac{1}{f} = \frac{1}{v} - \frac{1}{u} = -\frac{1}{100} + \frac{1}{25} = \frac{3}{100} \Rightarrow f = 33.3$ cm; $P = +3$ D	3
	OR: The eye could focus only at one distance; objects at other distances would appear blurred	3

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) distances from optical centre; along incident light +; heights above axis + (ii) $\frac{1}{u} = \frac{1}{v} - \frac{1}{f} = -\frac{1}{15} + \frac{1}{20} = -\frac{1}{60} \Rightarrow u = -60$ cm	2+3
	OR: (i) image at F behind mirror, highly diminished; image between P and F, diminished, virtual (ii) parallel rays diverge after reflection (iii) rear-view mirror	3+1+1
20	(i) dispersion, refraction and internal reflection in raindrops (ii) raindrops act as tiny prisms (iii) behind the observer	3+1+1
	OR: (i) redirection of light by particles (ii) very fine particles scatter blue; large particles scatter all colours (white) (iii) blue scattered by air; at sunset blue is scattered away over the long path	1+2+2

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